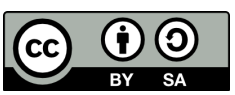
My First Bitcoin

Educator Guide
Version 2026.04

Bitcoin for Juniors

Preparing young minds to better understand Bitcoin



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Donate

Table of Contents

Foreword 6

FAQ 6

Introduction 8

1. Exchange, Value & Price

1.1 Ship Rekt 14

1.2 Going, Going, Gone! 22

1.3 Worth It 24

1.4 What is value? 28

2. Time, Energy & Reward

2.1 Work, Rest & Play 34

2.2 Energy Economy 38

2.3 That's Not Fair! 41

2.4 That's Not Fair (Story Time) 45

2.5 Time Preference 49

2.6 Smart Choices 52

3. Analog vs. Digital

3.1 Back In My Day 58

3.2 Vote With Your Feet 60

3.3 Invent The Future 63

4. Safety, Security & Privacy

4.1 Gatekeepers 68

4.2 Wire Tap 70

4.3 Cypherpunks Write Code 74

4.4 Nostr 77

5. Money Matters

5.1 Money Matters 82

6. Bitcoin Basics

6.1 From Problem to Protocol 88

6.2 Satoshi, Where Are You? 91

6.3 It's Mine! 94

Foreword

The world our children will inherit is being shaped right now. Money, information, and the tools of cooperation are evolving faster than most education systems can keep up. Whether the next generation meets that world with clarity and confidence, or confusion and dependency, depends largely on what we teach them today.

Education has always been more than the transfer of information. Its purpose is to develop independent thinkers who can question, evaluate, and reach their own conclusions. That capacity is more important now than ever.

My First Bitcoin exists to build that kind of education. We believe the current money system is broken and that Bitcoin represents sound money for a digital age. We believe education should be open-source and free from financial incentives. Empowering learners to think for themselves, not telling them what to think, is the work that matters most.

Bitcoin for Juniors sets a new standard for Bitcoin education for children. It is not a simplified version of our Bitcoin Diploma, but a reimagining. Through playful, age-appropriate, and interactive learning, children aged 8 to 12 explore core ideas like value, scarcity, exchange, fairness, privacy, and responsibility. It is not a technical course and does not promote any belief. It provides the building blocks for children to think and draw their own conclusions.

Educators are the multiplier. This program equips teachers and community educators with practical tools to deliver engaging, age-appropriate learning. The goal is not to create Bitcoin believers, but clearer thinkers who can question systems and participate in an open, decentralized world.

The next generation does not need to be told what to think about money. They need the foundation to figure it out themselves. Bitcoin for Juniors provides that foundation.

FAQ

1. What is Bitcoin for Juniors?

Bitcoin for Juniors is a foundational program for children aged 8 to 12. It introduces the core ideas required to understand Bitcoin, without requiring children to use it directly.

The program is delivered in two parts:

- A comprehensive Teacher Guide PDF with structured lesson plans, pedagogy notes, and facilitation guidance
- Printable student handouts and activity materials for interactive classroom use

It is a preparation course, not a technical training course.

2. Who is Bitcoin for Juniors for?

Bitcoin for Juniors is designed for:

- Classroom teachers
- Homeschool educators
- Community leaders and youth mentors
- Parents who want to introduce financial awareness early

Built for children aged 8 to 12, it is adaptable across different educational environments and cultural contexts.

3. Why do children need to learn about Bitcoin?

Children do not need to use Bitcoin wallets or manage private keys. What they need are the foundations to understand Bitcoin later.

B4J is intentionally low-tech — lessons use physical materials, not devices — making it accessible in any classroom environment and appropriate for younger learners. The goal is conceptual readiness, not technical mastery.

4. What does the program teach?

Bitcoin for Juniors teaches the key concepts required to understand Bitcoin, including:

- What money is and why societies use it
- The difference between value and price
- Time, energy, effort, and reward
- Scarcity and abundance
- Public versus private information
- Basic ideas behind encryption and codes
- The historical evolution of money
- The origin and purpose of Bitcoin

These ideas are explored through games, storytelling, movement, and collaborative activities. Children are not told what to believe about Bitcoin — they are given tools to think about money, value, and fairness, and reach their own conclusions.

5. Is this a course about investing, trading, or using real money?

No. This program does not teach children to invest, trade, or speculate. It does not involve real money. There is no financial risk and no promotion of investment behavior.

6. How is Bitcoin explained in an age-appropriate way?

Bitcoin is introduced through analogy, play, and guided discussion. Complex technical ideas such as cryptography or decentralized networks are simplified into concepts about trust, privacy, fairness, and responsibility. The focus is on understanding principles, not memorizing technical detail.

7. How can I teach this in my classroom or learning environment?

The program is designed to be low-tech and highly interactive. Lessons can be delivered using:

- Printed worksheets
- Role-play simulations
- Physical tokens, cards, or simple classroom materials
- Guided group discussion

The Teacher Guide provides step-by-step instructions, classroom management tips, reflection prompts, and extension ideas. No advanced technical knowledge of Bitcoin is required.

8. What happens after students complete the program?

Students leave with stronger financial awareness, critical thinking skills, and a structured understanding of money's evolution. They are not "Bitcoin users." They are prepared learners. When they encounter Bitcoin later in life, they will have the intellectual foundation to understand it responsibly and independently.

Introduction

This guide explains how to read and use the lesson plans in Bitcoin for Juniors. It walks through each section of the lesson plan format so educators know what to expect, how to prepare, and how to adapt the content for their context. The structure is here to serve you — not the other way around.

Using the Lesson Plans

Each lesson plan in Bitcoin for Juniors is a structured guideline. It is not a script that must be followed word for word. Teachers are encouraged to adapt, simplify, expand, or modify activities based on their students, setting, and time constraints. The plans are designed to support clarity and confidence, especially for educators who are new to teaching money or Bitcoin concepts.

Adjustments are expected. Facilitate rather than lecture. The framework provides structure. The teacher provides judgment and context.

Understanding the Structure

Each activity follows a consistent format. This allows teachers to quickly identify what is needed and how to run the session effectively.

Activity Type

Activities are categorized for ease of planning. For example: simulation game, guided discussion, experiential learning, cooperative challenge. These labels help teachers select activities that match their instructional goals and classroom energy levels.

Duration

The time listed is approximate. Some groups will move quickly. Others will require more discussion. Teachers should treat duration as a flexible guide rather than a strict limit.

Grouping

Indicates whether the activity is designed for:

- Whole class
- Pairs
- Individual work

Grouping helps establish the intended learning dynamic. Working in groups tends to increase engagement and helps students reinforce ideas through discussion.

Learning Outcomes

Learning outcomes define what students should know, understand, or be able to do by the end of the activity. These outcomes are essential. They help the teacher remain focused on purpose rather than simply completing a task. If time is limited, prioritize the learning outcome over finishing every step.

Materials vs Equipment

Materials

Materials are consumable or hands-on items used directly by students. Examples: worksheets, printed cards, pencils, beads, sticky notes, posters. These are items students manipulate during the activity.

Equipment

Equipment refers to non-consumable tools or environmental needs. Examples: whiteboard, timer, projector, open classroom space. Equipment supports the activity but is not handled directly by students in the same way materials are.

The distinction helps teachers prepare efficiently.

PROCEDURE

Each lesson includes a procedural sequence to guide flow and pacing.

OPENING

Completed before students enter the room. This includes preparing materials, arranging seating, writing prompts on the board, and organizing resources. Preparation is best done without students present to ensure smooth transitions.

PRE-ACTIVITY

The warm-up stage. This sets expectations and sparks curiosity. It may include a question, short story, scenario, or challenge. The goal is to activate prior knowledge and prepare students for the task.

Objective in Student's Words

The teacher briefly explains the purpose of the activity using language students understand. When students know why they are doing something, engagement and retention increase.

ACTIVITY

1. **Setup** — Instructions for organizing students and distributing materials.
2. **Model** — When appropriate, demonstrate the task. Modeling reduces confusion and builds confidence.
3. **Carry Out** — Students complete the activity. The teacher observes, guides, and supports rather than dominates.
4. **Checkpoint** — Pause briefly to confirm understanding. Adjust if needed. This prevents misunderstandings from escalating.
5. **Reflection** — Students discuss what happened and what it means. Reflection is critical. It transforms experience into learning. Encourage open-ended questions and connections to real-world concepts.

FOLLOW-UP

A short wrap-up moment. Reinforce the key idea. End positively. This may include a recap question, quick partner share, or a brief statement connecting the activity to the broader chapter theme.

CLOSE

Notes for the teacher. This section includes reminders to collect materials, store resources, reset the classroom, and prepare for the next session.

Classroom Management

Each classroom differs in age range, energy level, and group dynamics. Suggestions provided in this section may include:

- Clear instructions before movement
- Defined time limits

- Visible rules
- Structured roles within groups

Teachers should adapt management strategies to their context.

Extensions & Sponge Activities

If time remains, extension ideas — sometimes called sponge activities — allow the activity to continue with minimal preparation. These may deepen discussion, add complexity, or introduce a creative element. They are optional and flexible.

Differentiation

Differentiation ensures accessibility for diverse learners. Adjustments may include:

- Simplified language for younger students
- Visual supports for emerging readers
- Clear step-by-step instructions
- Modified pacing
- Alternative ways to respond

Accessibility may also involve physical safety considerations and inclusive participation structures.

APPENDIX

The appendix contains supporting materials, templates, printable resources, and links referenced in the lesson. Teachers should review the appendix before class to ensure all necessary materials are prepared.

Final Guidance

These lesson plans are structured for clarity, but flexibility is expected. Focus on the learning outcomes. Allow space for student voice. Prioritize interaction over perfection.

Bitcoin for Juniors is designed to support confident teaching and meaningful learning.

Module 01

Exchange, Value & Price

1.1 Ship Rekt

1.2 Going, Going, Gone!

1.3 Worth It

1.4 What is value?

1.1 Ship Rekt

Activity type

- Simulation - Roleplay - Game

Duration

- 60 minutes

Grouping

- Whole class

Description

In this 60 minute whole class simulation, students roleplay as shipwrecked survivors who must secure essential resources to “survive” a week on a desert island. Using playing cards to represent tools and supplies, learners begin with pure barter, practicing negotiation while encountering common trade barriers such as surplus, scarcity, and the coincidence of wants. A timed “Shell Hunt” then introduces commodity money, as students collect tokens that can be used to simplify exchange. With shells in circulation, the class enters a guided marketplace where students set prices, respond to supply and demand, and experience price discovery. The activity culminates in budgeting and a structured auction or trade round for “treasure” wishlist items, bridging to why money exists.

Learning Outcomes

- Students experience barter, the “coincidence of wants,” scarcity, supply/demand, price discovery, budgeting, and auctions.
- Bridge to money: commodity money (shells) and why money helps trade.

Materials

- Playing cards
- “Shells” (physical tokens: paper circles, buttons, craft shells)
- Treasure chest (Small box/enveloppe)
- Gift card (expired or fake $100 gift card)
- Blank flash cards (or slips of paper)
- **Nice-to-have:** Visual aids (*See Appendix A & B*), Treasure map, Reward stickers

Equipment

- Timer/phone stopwatch
- Whiteboard or screen
- Marker pen
- **Nice-to-have**: Whiteboard/markers, projector, bell/signal for time, music (optional for treasure hunt)

PROCEDURE

OPENING

3 minutes

- **Hide** the commodity money (shells) around the classroom before students enter the room
- **Review** vocabulary as a warm-up: Trade, money, price... etc
- **Review** call-back signals and time rules.

Objective In Student's Words

- Trade objects to survive a week on the desert island.

PRE-ACTIVITY

5 minutes

1. **Narrate** a back-story explaining how students are shipwrecked. Use visual aids if necessary. (*See Appendix A*)

For example...

"You live in a small coastal town and work as a fisherman. One day, you head out to sea with your crew in search of the day's catch. All is calm and the boat gently chugs along a glassy ocean with gulls flying overhead. As you lose sight of the shore and the boat heads into deeper and deeper water, there on the horizon, a storm is brewing. You see dark clouds gathering and heading towards the boat. The captain decides to head to shore and makes a u-turn towards port, but it's too late. The storm is closing in fast and it's chasing the crew faster than the boat can carry them - there's no escape.

Within minutes the ocean's swell is breaking over the bow of the ship. The wind is whirling, there's spray whipping through the air and the boat starts to rock violently from side-to-side. As you look over your shoulder, you see a giant wall of dark clouds heading from the ocean up to the heavens - the storm is coming for you and the crew..."

1. **Ask students** to name objects they would need on the desert island to survive. **Elicit** the four objects chosen for the activity (i.e. a machete, a fishing rod, a tent and matches) and show them to the class using *visual aids*.
2. **Explain** that the *playing cards* will represent the object they need to survive.
3. **Rephrase** the objective and purpose in the student's words.

DAY 1 - BARTER

10 minutes

Setup

1. **Distribute** several cards to each student several cards of the same suit.

- *Student A = Diamonds, Student B = Hearts etc.*

1. **Explain** that the suit of the card represents the objects they have found

- *Diamond = Machetes, Hearts = Fishing rods, Clubs - Tents, Spades - matches*

1. **Ask** each student what they would do with that object. When they explain how they would use their object, **take a card** from the student and place it on a 'used' pile in the middle of the group.
2. Once each student has used one of their objects, **ask** what they will do with the surplus objects - **elicit** that they will have to trade what they HAVE for what they NEED (See *Appendix B*)

Model

- Teacher demos a simple trade with a student (offer ↔ counteroffer ↔ deal). Write sentence frames on board, if required:"I have ____. I need ____." / "I can trade ____ for ____."

Trade

- Explain the trade grid (See *Appendix B*)
- Students have **5 minutes** to barter what they HAVE for what they NEED.
- When the time is up, use call-back to end.

Checkpoint & Debrief

- Thumbsup : met all needs / almost / not yet
- Discuss: What was easy/hard? Who had too much? Too little? Introduce **scarcity** and **supply & demand**.

Reflection

- Prompt students: "One thing that helped me trade was _____."

DAY 2 - BARTER... again

10 minutes

Setup

1. **Review** the suits of the cards and the objects they represent

- Diamond = Machetes, Hearts = Fishing rods, Clubs - Tents, Spades - matches

1. Once again, **ask** each student what they would do with their newly acquired object. When they explain how they would use their object, take a card from the student and place it on a 'used' pile in the middle of the group.
2. Once each student has used one of their objects, **ask** what they will do with the surplus objects - **elicit** that they will have to trade once again what they HAVE for what they NEED.

Model

- Teacher demos a simple trade with a student (offer ↔ counteroffer ↔ deal). Write sentence frames on board, if required:"I have ____. I need ____." / "I can trade ____ for ____."

Trade

- Explain the trade grid (*Appendix B*)
- Students have **5 minutes** to barter what they HAVE for what they NEED.
- When the time is up, use call-back to end.

Checkpoint & Debrief

- Thumbsup : met all needs / almost / not yet
- **Discuss**: What was easy/hard? Who had too much? Too little?

Mini-Lesson

- Day 2 should be harder than Day 1 of trading.
- Students quickly discover it's **hard** unless two people want what the other has (**coincidence of wants**).
- Name the problem: "I want A, you have B... trading is stuck."
- Bridge: "People invented **money** to fix this problem."

DAY 3 - SCARCITY & EFFORT: "Shell Hunt"

10 minutes

Setup

- If the tokens (shells) haven't already been hidden around the classroom, take some time to complete this now.
- **Explain**: "Shells are special on this island. They can be used as money. Collect as many as you can."
- **Establish** rules of respect and safety - no stealing, no pushing etc.

Treasure Hunt

- 5-minute **shell hunt** (hide 2–3 times class size around room/hall; vary difficulty).
- Remember to count how many shells you hide around the room.

Count & Record

- Teams/pairs count shells. Quick bar graph on board (who has many/few).
- Discuss **scarcity** and **effort**.

Mini-Lesson

- Define **commodity money** (a thing people use as money).
- Use *visual aides* to **illustrate** examples of money in the past.
- **Ask**: "Could shells be money here?"

Reflection

- "If shells become money, what might the **price** of items be? Why?"

DAY 4 - COMMODITY MONEY & PRICES: "Shell Market"

10 minutes

Setup

1. **Review** the suits of the cards and the objects they represent

- *Diamond = Machetes, Hearts = Fishing rods, Clubs = Tents, Spades = matches*

1. **Ask** each student what they would do with their newly acquired object. **Listen** to their usage and **discard** a card to the 'used' pile.
2. **Elicit** that they will have to trade once again what they HAVE for what they NEED but this time they can also use the commodity money - shells.

Model

- Teacher demos a simple transaction with a student (offer ↔ counteroffer ↔ deal). Write sentence frames on board, if required:
  "I have ____. I need ____." / "I can trade ____ for ____."
- Emphasise that the students can decide on the price of the items

Trade

- Explain the trade grid (*Appendix B*)
- Students have **5 minutes** to trade what they HAVE for what they NEED.
- When the time is up, use call-back to end.

Checkpoint & Debrief

- Thumbsup : met all needs / almost / not yet
- **Discuss**: *What was easier/harder with money? What prices did you pay? Did you end up with more or less shells? Was this fair?*

DAY 5 - BUDGETS & AUCTIONS: Treasure Chest

15 minutes

Prep

- **Hide** a treasure chest somewhere in the classroom when students are completing the Day 4 trade.

Story Hook

- The stranded crew look around and find some treasure
- Include a $100 Amazon/Walmart etc. gift card (fake)

Treasure hunt

- Student search the desert island (classroom) for the treasure chest
- Once it has been found, explain the bounty and how it will be used.

Wishlist

- Students can use the giftcard to place one luxury item each on to the wish list.
- Take a pen and paper and list each item requested by the students and write their name next to the item.
- The order is placed and the next day, the items arrive.

DAY 6 - WISHLIST

For this penultimate activity there are two options to carry out:

OPTION 1 – Random distribution and Trade (SIMPLE)

The wishlist items can be randomly distributed to the students. This can be explained through a creative story. For example, there is no landing strip on the island, so the items were airdropped by parachute and strewn across the island. From this point, students can trade the luxury item that they HAVE for the item they WANT. They can use their remaining shells to find what they want.

Trade Prep

- Explain to students, that as when trading the playing cards, they can decide on price and trade what they HAVE for when they NEED.

Trade

- Students have **5 minutes** to trade what they HAVE for what they NEED.
- The objective is to secure the luxury item they requested.
- Shells and other items (playing cards) can be included in trades.
- When the time is up, use call-back to end.

OPTION 2 – Auction (ADVANCED)

- The person who found the gift card shipped the items to their shelter on the island. They now own all the items and can sell them off to the highest bidder.
- The teacher now sells items back via **auction**. Students use their shells to compete in the auction and try and win their luxury item.

Auction Prep (5 min)

- Explain an **English auction** (prices go up; last hand wins). Practice paddle up/down.
- Budget rule: never bid more shells than you have; plan ahead.

Auctions (10 min)

- Auction items one by one to students.
- Students bid on the items using their shells.

Auction Debrief (5 min)

- What is a **budget**? How did prices show **demand**? Was it fair? How could rules change outcomes?

FOLLOW-UP (2 min)

- What items were **expensive**? Why? Where did **supply** feel low? What strategies worked?
- "Money made trading easier because _____."
- Barter was hard → shells helped (commodity money) → money solves coincidence of wants.
- Bridge to Bitcoin – discuss that Bitcoin is a money that helps people exchange goods and services.
- Discuss some of the differences between Bitcoin and the shells they were using.

DAY 7 - FREEDOM

- Using all of the objects they have, see if they can create a way off the island.
- Ask students to THINK-PAIR-SHARE to find the key ideas.
- As a class, formulate a plan to escape the island and find their way back to safety.

FOLLOW-UP

- Quick write or drawing: "One thing I learned about **trade** or **money**."
- Gallery walk of price tags & auction results; discuss fairness and markets.

- Optional challenge (older kids): Design a new island money (rules, supply cap, how to prevent cheating).

CLOSE

- Gather all materials
- Verify all playing cards have been returned to ensure equal distribution between all suites for another activity.

NOTES

Classroom management

- Set “Marketplace” boundaries; use a call-back (eg. “Bit, block… BOOM!”).
- Trading rules posted: 1) Be kind, 2) Offer/Counteroffer, 3) No grabbing, 4) Finalize by handshake/high-five, 5) Return to places on call-back.
- If required, assign roles (Banker/Pirate helpers, Market Monitors) to channel energy and support younger learners.

Extensions & Sponge Activities

- **Draw your island store** and price three items in shells.
- **Mental math**: If you have 12 shells and win an auction for 9 shells, what’s left?
- Make a **venn diagram** to show what kids value at home and on the island - are there objects which fall into both?
- **Top 10 Wishlist** - ask students to write a list of the top 10 items they would pack if they were going to a desert island.

Differentiation

- Younger students trade fewer categories (just Food/Water); older students handle more complex needs and dynamic pricing.
- ELL/Accessibility: Provide picture cards; sentence frames for offers; assign peer buddies.
- Safety: Movement is walking only; clear boundaries; timed rounds.

APPENDIX

Appendix A - Story visual aids

Appendix B - Trade Table

Appendices

- Appendix A - Story visual aides
- Appendix B - Trade Table

1.2 Going, Going, Gone!

Activity type Role-play • Auction

Duration 45 minutes

Grouping Whole Class

Description

In this activity, students take part in a classroom auction using "shells" (or tokens) as a simple form of money. The class runs a live auction where everyday objects are sold one at a time to the highest bidder. As prices emerge in real time, students observe how scarcity (limited items and limited money) and demand (how many people want the same item) shape what something "costs." A guided reflection links their auction outcomes to core ideas in value, exchange, and price formation, emphasizing that prices are not fixed and that different people value the same object differently.

Learning Outcomes

- Students experience scarcity and price formation by participating in a live auction.
- They learn that objects gain value depending on how many people want them, and how limited they are.
- Students understand that:
  - Scarcity makes items more valuable.
  - Prices are not fixed; they depend on demand.
  - People value different things differently.

Materials

- 10 small classroom objects (fun or useful: stickers, pencils, snacks, erasers, funny hats, toys, etc.)
- Shells (or paper tokens/coins if no shells available) — at least 3–5 per student

Equipment

- Pen & paper for recording bids
- **Nice-to-have**: Whiteboard and markers, Auction bell or clap/signal

PROCEDURE

OPENING

2 minutes

- Predetermine how many tokens each student is to receive.
- Arrange 10 items on a table at the front (the "auction house")

Objective In Student's Words

- Use shells to bid in an auction and see how prices change when lots of people want the same thing and there aren't many of them.

PRE-ACTIVITY

5 minutes

1. **Review** the items on the table with students. Discuss names and functions of the items, if required.
2. **Announce**: "These shells are your money. You'll use them to buy things in today's game.
3. **Distribute** the shells and get students to count how many they possess.

ACTIVITY

30 minutes

Setup

1. **Explain** that the items on the table will be sold one by one.
2. Students will **bid** using shells — whoever bids the most wins the item.

Model

1. **Pick one student** as the "bidder." Hold up an item, name it clearly, and start the opening bid at 1 shell.
2. **Model bidding**: ask the student to raise their hand, say their bid out loud, and place the shells where everyone can see them. Repeat once by "outbidding" them yourself (or with a second student) to show how bids increase.
3. **Close the sale**: do a clear countdown ("Going once, going twice, sold"), ring/clap, announce the final price, hand over the item, and record the price visibly.

Live Auction

1. The teacher holds up an item.
2. Start at 1 shell, let students bid higher.
3. Record the final price on a sticky note and stick it to the item (visible to all).
4. Repeat for each of the 10 items.

Checkpoints

- Throughout the auction, remind students to be aware of their remaining shells.
- Explain that once they have no more shells, they can not bid on any further items.
- Students need to 'budget' for their items.

FOLLOW-UP

1. Lay all items with their final prices on a board/wall.
2. Ask:
    1. Which items were most expensive? Why?
    2. Were some things not worth much? Why?
    3. Did everyone want the same things?
    4. Did anyone run out of shells too quickly?
3. Highlight: Price comes from value *and* scarcity.
4. Define these terms and write them on the board.

CLOSE

- Distribute the items to the winning bidders and congratulate them

NOTES

Classroom management

- Some students may find this concept unfair. They may have been outbid on items they wanted. Be aware that this could be a negative learning experience. Offer the possibility of further trade and barter (see *Differentiation* below)

Differentiation

- Modulating the distribution of tokens can change the outcome of the activity. The simplest, and fairest method is to distribute the tokens equally. However, this is not the most realistic situation. Unequal distribution may be suitable with more advanced students to discuss financial mobility and inequality.
- Younger students: If bidding is not easy for students, it is possible to ask around the group in turn.
- ELL/Accessibility: Children can indicate their bid by placing shells at the centre of the table, poker-style.

Extensions & Sponge Activities

- If no actual giveaway: instead of giving the items away, students write on post-its what they think each item *should* cost. Compare their "predictions" with the actual auction results.
- This makes the activity more theoretical and allows for repeating without needing prizes.
- Time permitting, a trade market can be opened up with students who are not satisfied with the auction. Students can be encouraged to barter and use any remaining shells to trade with each other. This can either be structured and led via the teacher, or decentralized and sporadic, allowing students to take the initiative.

1.3 Worth It

Activity type Sorting • Visual classification • Guided discussion

Duration 45 minutes

Grouping Whole class with individual turns

Description

Students explore the meaning of price by cutting out everyday items and placing them along a shared price line from cheap to expensive. Through comparison, discussion, and teacher guidance, learners develop a concrete understanding that price is the amount of money something costs and that prices can vary widely between items.

Learning Outcomes

By the end of this activity, learners will:

- Understand that price means how much something costs in money
- Recognize the difference between cheap and expensive items
- Compare prices across different types of goods
- Participate in group discussion and justification

Materials

- Old catalogues, magazines, or printed product sheets
- Glue sticks or tape
- Markers
- **Nice-to-have:** Sticky notes or voting stickers

Equipment

- Child-safe scissors
- **Nice-to-have:** Projector, Visual Aides (*See Appendix A*)

PROCEDURE

OPENING

10 minutes

1. Optional: Pre-select and cut out items for modelling the activity
2. Ensure all contents of the catalog are age appropriate.
3. If possible, attempt to hide the retail price of the items using a marker pen.

PRE-ACTIVITY

5 minutes

1. Ask students: “Who knows what the word *price* means?”
2. Write clearly on the board:
    1. **Price = how much money something costs**
3. Show two contrasting examples (e.g., apple vs bicycle) and ask which costs more.

Objective In Student’s Words

- “Learn how much things cost and tell the difference between cheap and expensive.”

ACTIVITY

25 minutes

Setup

1. Create a long horizontal line on the wall or board using tape or a dry-wipe pen. (*See Appendix B*)
2. Label:
    1. Left: “Low Price”
    2. Right: “High Price”

Model

1. If not already done, select and cut out and item from the materials
2. Place one item yourself on the horizontal line and explain your thinking aloud:
    1. “This goes here because it has a low price - it usually costs less money.”
3. Ask students if they agree or disagree.

Carry Out

1. Students take turns coming to the front and placing their items on the price line.
2. Teacher prompts:
    1. “Is this price lower or higher than the last one?” ir “Does it cost more money to buy?”
    2. “Should it go more to the left or right?”
3. Encourage movement and adjustment as opinions change.

Reflection

- Walk along the price line as a group.
- Ask:
    - “Do you think this item should move?”
    - “Which item surprised you?”
- Reveal or estimate real prices for a few items and compare them to student guesses.

- Highlight:
  - Small items can be expensive
  - Big items can sometimes be cheap

FOLLOW-UP

2 minutes

Summarize:

- Price helps us compare items
- Prices are different for different things
- We can disagree, but money helps us decide

CLOSE

1. Congratulate students for their participation.
2. Take a moment for gratitude for the objects in their life.
3. Ask students to return all materials and equipment.
4. The project can be displayed or stored and can be added to later by the students.

NOTES

Classroom management

- Call students up one at a time
- Encourage respectful disagreement

Extensions & Sponge Activities

- Add price categories (Under $10, $10–$50, Over $100, Over $1000)
- Redo the line using food, experiences, or services
- Compare guessed prices vs real prices
- Higher or Lower game
- Introduce scarcity by asking why some items cost so much

Differentiation

- **Younger students:** pre-cut items
- **ELL/Accessibility:** allow pointing instead of speaking
- **Safety:** supervise scissor use

Appendices

- **Appendix A - Visual Aid**
- **Appendix B - High Price / Low Price Visual**

1.4 What is value?

Activity type Discussion • Collaborative art project

Duration 45 minutes

Grouping Whole class with individual contributions

Description

Students explore the concept of value by identifying everyday items from their own lives and positioning them on a shared visual chart. Through guided discussion and a collaborative wall-based activity, learners compare personal value with price and reflect on how value can change depending on perspective. The activity builds foundational understanding of value, price, and scarcity through discussion, movement, and visual thinking.

Learning Outcomes

By the end of this activity, learners will:

- Understand that *value* is based on personal and social needs.
- Recognize the relationship between *price* and *value*.
- See that value is not fixed but changes depending on perspective.
- Practice collaboration through shared discussion and group art.

Materials

- Sticky notes or index cards
- Markers, pens or pencils
- Tape or sticky putty
- Low-tack tape (masking tape, washi tape, painters tape etc)
- **Nice-to-have:** A selection objects from home

Equipment

- **Nice-to-have**: Whiteboard & markers

PROCEDURE

OPENING

10 minutes

1. Find a location in the classroom suitable for a large art display. Ideally a flat, smooth surface.
2. Clear the wall of any objects and clean the wall to ensure sticky notes can adhere to the wall.
3. Place any objects from your home on the table at the front of the class before students enter.

Note: Using a whiteboard at the front of the class can be suitable for the activity. However, for maximum impact, it is best to carry out this activity in a separate location in the classroom so the art project can be displayed for a longer period of time.

PRE-ACTIVITY

5 minutes

1. Allow students to identify objects you have placed in front of them.
2. Discuss their price and value.
3. Introduce the two ideas:
    1. **Price**: how much something costs
    2. **Value**: how useful or important something is
4. Ask students to give their opinion on the object and if they agree or disagree.

Objective In Student's Words

- "Find out what decides the price of things."
- "Learn what things are valuable to us and why not everyone agrees."
- "Talk about why expensive things aren't always valuable, and cheap things can still matter."

ACTIVITY

30-40 minutes

Setup

1. Ask students to list as many items as they can think of from their bedrooms or houses (toys, clothes, gadgets, books, etc.).
2. Each student writes one item per sticky note.
    1. Prompt questions:
        1. "Which things are important to you?"
        2. "Which things don't really matter?"
3. If possible, each student can have their own respective color of sticky note to easily distinguish items later in the activity.
4. Tape a long horizontal line across the board (*See Appendix B*).Label this axis:
    1. Left = "Low Price"
    2. Right = "High Price"
5. Then add a vertical line crossing the center. Label this axis:
    1. Bottom: "Low value"
    2. Top: "High value"

Model

1. Take one of the items from home and write the name of the item on a sticky note.
2. Demonstrate where to place it within the four quadrants.

Carry out

1. Students place their sticky notes along the axis where they think their item belongs.
2. Support students by encouraging them to add as many items as possible. If students have added all of their sticky notes, prompt them for new items.

Reflection

- Guiding questions:
  - “Is this cheap or expensive?”
  - “Would most people agree on the price?”
  - “How useful is this?”
  - “Would everyone value this the same way?”
- Prompt discussion:
  - “Which items surprised you?”
  - “Why do some cheap things have high value?”
  - “Why do some expensive things feel useless?”
  - Optional perspective shift:
- Optional perspective shift:
  - Have students review another student’s items and discuss differences.
- Key takeaway:
  - Price is a number
  - Value is a judgment

FOLLOW-UP

- Review:
  - Price and value are not the same
  - People disagree on value
  - Prices help people trade even when they disagree
- Bridge to money:
  - Money and prices help us trade when value is subjective

CLOSE

1. Identify any trends or patterns.
2. Congratulate students for their participation.
3. Take a moment for gratitude for the people, times and objects in their life.
4. Ask students to return all materials and equipment.
5. The project can be displayed or stored and can be added to later by the students.

NOTES

Classroom management

- Encourage respectful disagreement
- Reinforce that there are no wrong answers
- It can be more engaging to tape out and label the axes during the activity. This could also distract focus and lose concentration if the process is complicated.

Extensions & Sponge Activities

- Show and Tell
- Add a third layer: “Who decides the price?”
- Re-introduce auctions or trading games

Differentiation

- **Younger students**: pre-printed item cards
- **ELL/Accessibility**: drawing instead of writing
- **Safety**: supervise movement near walls

Appendices

- **Appendix A - Visual Aid**
- **Appendix B - High Price/Low Price Visual**

Module 02

Time, Energy & Reward

2.1 Work, Rest & Play

2.2 Energy Economy

2.3 That's Not Fair!

2.4 That's Not Fair (Story Time)

2.5 Time Preference

2.6 Smart Choices

2.1 Work, Rest & Play

Activity type Interactive group activity - Personal reflection

Duration 45 minutes

Grouping Whole class for collaborative activities, then individual or pairs for reflection

Description

Students explore how everyday activities can be grouped into work, rest, and play. Through a shared classification activity and individual routine mapping, learners reflect on how they use their time across a day. The activity builds early awareness of balance, energy, and intentional time use.

Learning Outcomes

By the end of this activity, learners will:

- Identify and classify activities as work, rest, or play
- Reflect on how their time is distributed across daily life
- Recognize the importance of balance between work, rest, and play
- Begin connecting time use to energy and well-being

Materials

- Flashcards with common daily activities (e.g. brushing teeth, homework, soccer, video games, sleeping, cooking, reading)
- Blank routine handouts:
- Pencils, crayons, markers
- **Nice-to-have:** Extra blank cards for student-created activities, Visual Aids (*See Appendix A*)

Equipment

- Large poster or board space divided into WORK – REST – PLAY (*See Appendix B*)
- Whiteboard or wall space
- Tape or magnets for attaching cards
- **Nice-to-have**: Timer or visual clock

PROCEDURE

OPENING

3 minutes

1.

PRE-ACTIVITY

5-10 minutes

1. Ask students:

- "What is work?"
- "What is rest?"
- "What is play?"

1. Collect a few responses without correcting them. Explain that some activities may fit more than one category. The Visual Aids can be used to support this step (*See Appendix A*)
2. Explain the three categories on the board:

- Work: things we must do
- Rest: things that help us recover
- Play: things we choose for fun

Objective In Student's Words

- "Explore out what I do for work, rest, and play, and see if I have a good balance."

ACTIVITY

30 minutes

Setup

1. Prepare the poster or board with three clear overlapping sections (*See Appendix B*):

- WORK
- REST
- PLAY

1. Place the activity flashcards where all students can see them.

Model

1. Choose one card (e.g. "sleeping" or "homework") and think aloud:

- "I might put this in 'REST' because it helps my body recover."

1. Place the card
2. Ask students if they agree or would place it somewhere else.

Round One: Collaborative Classification

1. Students take turns selecting a flashcard and placing it under WORK, REST, or PLAY.

2. Teacher prompts:

- "Why did you choose that category?"
- "Could this fit somewhere else?"

1. Optional variation:

- Students vote before placement
- Memory-style matching before classification

Checkpoint

1. Review the completed poster together.
2. Ask:

- "Does anything belong in more than one group?"
- "Do we all agree?"

1. Reinforce that categories can overlap.

Round Two: My Routine (20 minutes)

1. Students complete one or more routine handouts:

- Daily routine
- Weekly routine
- Yearly routine

1. They fill in activities from their own lives.
2. Discussion prompts:

- "Where do you see work in your routine?"
- "When do you rest?"
- "What play happens every day, week, or year?"

Reflection

Optional deeper discussion:

- "What happens if there is too much work?"
- "What happens if there is no rest or play?"

FOLLOW-UP

5 minutes

1. Summarize key ideas:

- Everyone needs work, rest, and play
- Balance matters
- Different people have different routines

1. Connect forward:

- These choices affect energy, health, and learning.

1. Optional home extension: Students ask a parent or caregiver what work, rest, and play looked like when they were the same age.

CLOSE

1. Collect flashcards and handouts
2. Remove or clear the poster if needed
3. Thank students for participating
4. Final wrap-up message to students: "Everyone's routine looks different, but balance matters."

NOTES

Classroom management

- Encourage respectful disagreement
- Allow students to change their minds

Extensions & Sponge Activities

- Redo the activity for weekends vs school days
- Compare routines across seasons and holidays

Differentiation

- Older students: Students can write their own cards using sticky notes.
- ELL/Accessibility: use icons and gestures
- Safety: supervise movement around the board

APPENDIX

Appendix A - Visual Aid

Appendix B - Venn Diagram

Appendices

- **Appendix A - Visual Aid**
- **Appendix B - Venn Diagram**

2.2 Energy Economy

Activity type Interactive reflection • Graphing activity

Duration 35 minutes

Grouping Whole class discussion, then individual graphing

Description

Students explore how their energy changes throughout the day by plotting common activities on a shared class graph and a personal Energy Tracker. By visualizing energy highs and lows across time, learners build early self-awareness about effort, rest, and how energy is spent. This activity lays the foundation for later lessons on time preference, value, and intentional decision-making.

Learning Outcomes

By the end of this activity, learners will:

- Recognize that energy levels change throughout the day
- Identify activities that give or use energy
- Visualize personal daily energy patterns
- Begin reflecting on how time and energy are used

Materials

- Flashcards or image cards of daily activities (eating breakfast, brushing teeth, walking to school, homework, playing outside, watching TV, sleeping, etc.)
- Individual Energy Tracker handouts (*See Appendix B*)
- Crayons, pencils, markers
- **Nice-to-have:** Printed mini images for non-writers, Visual Aids (*See Appendix A*)

Equipment

- Large whiteboard or poster for class Energy Tracker graph
- Whiteboard markers
- Tape or magnets for flashcards
- **Nice-to-have**: Projector or visual timer

PROCEDURE

OPENING

1. Prepare a large Energy Tracker graph on the board or poster:
2. Horizontal axis: Time of Day (Sunrise → Sunset)
3. Vertical axis: Energy Level (Low → High)
4. Lay out flashcards so they are visible and easy to access.
5. Prepare individual handouts and writing materials.
6. Ensure clear space for students to approach the board safely.

PRE-ACTIVITY

3-5 minutes

1. Introduce the idea:
2. Ask the class:
3. Make it personal:
4. Use quick show-of-hands questions:

Objective In Student's Words

- "See how my energy changes during the day and what uses or gives me energy."

ACTIVITY

30 minutes

Setup

1. If possible, use visual aids to illustrate time and energy tracking (*See Appendix A*)
2. Return to the Energy Tracker graph previously drawn.
3. Explain that the horizontal line shows the day from sunrise to sunset.
4. Explain that the vertical line shows energy from low at the bottom to high at the top.
5. Ask students to name examples of activities they do in the morning, afternoon, and evening.

Model

1. Choose one flashcard such as eating breakfast.
2. Ask: "When do you usually do this?" and "Does this give you energy or take energy?"
3. Place the card on the graph using student input.

1. Group Graphing

15 minutes

1. Show one activity card at a time.

2. Ask: "When do you usually do this?" and "Does this give energy or take energy?"
3. Use class consensus to place the card left to right based on time of day and up or down based on energy level.
4. Repeat with 8–10 activity cards.
5. Encourage discussion and disagreement by reminding students that activities can feel different for different people.
6. Step back and review the completed graph.
7. Ask: "What do you notice?" "When does energy seem highest?" "What happens later in the day?"

2. Individual Graphing

10–15 minutes

1. **Distribute** Energy Tracker handouts. (*See Appendix B*)
2. **Instruct** students to think about a typical day, choose 4–6 activities they usually do, and plot them on their own graph using words or drawings.
3. Support non-writers by providing printed images to glue or tape, allowing drawing instead of writing, and pairing with a buddy if needed.

Reflection

2 minutes

1. Ask: "What time of day do you have the most energy?" "Which activity uses the most energy?" "What helps you feel more rested?" "How could you use your energy better?"
2. For older students ask: "What happens if all your energy is used in the morning?" and "How can we save energy for things that matter?"

FOLLOW-UP

3 minutes

Content review with students

1. Review that energy changes during the day, activities affect energy differently, and people have different energy patterns.
2. Connect forward by explaining that these patterns affect how we work, rest, play, and make choices.

CLOSE

5 minutes

1. Collect handouts and materials.
2. Remove or clear the class graph if needed.

3. Thank students for participating and reinforce the message that energy is limited and how we use it matters.

NOTES

Classroom management

- Use clear turn-taking rules, limit placement decisions to a few seconds, encourage listening and respectful disagreement.

Extensions & Sponge Activities

- Compare school day vs weekend energy graphs, small-group graph comparisons, physical energy demo using crouch for low energy and stand tall for high energy.
- Use the same principles to look at energy across longer time scales (weekly, seasonally, lifespan etc)

Differentiation

- Younger students: Optional partner support for younger students, drawing and image cards.
- ELL/Accessibility: visuals, modeling, partner support.
- Safety: supervise movement near the board.

APPENDIX

Appendix A - Visual Aid

Appendix B - Energy tracker handouts

Appendix C - Activity Planner

Appendices

- Appendix A - Visual Aid
- Appendix B - Energy tracker handouts
- Appendix C - Activity Planner

2.3 That's Not Fair!

Activity type Game-based exploration – Structured discussion – Experiential learning

Duration 60 minutes

Grouping Teacher-led demonstration, individual participation, whole class reflection, optional pairs or small groups

Description

Students explore the idea of fairness by participating in a sequence of short games that distribute rewards in different ways. Through direct experience, learners compare unfair systems, random systems, effort-based systems, and Proof of Work. Emotional responses are welcomed and used to support structured reflection. The activity builds a foundational understanding of fairness, reward, and equal opportunity.

Learning Outcomes

By the end of this activity, learners will:

- Experience and identify unfair, random, effort-based, and fair systems
- Understand that not all rewards are earned equally
- Recognize that fairness requires equal rules and equal opportunity
- Build vocabulary and emotional awareness around effort, reward, and fairness
- Prepare conceptually for Proof of Work

Materials

- Beads or small tokens to represent rewards (“My Stack”)
- Flashcards or images showing qualities or attributes (hair colour, clothing, birthdays, etc.)
- Drawing paper and pencils
- Dice
- Deck of playing cards
- **Nice-to-have:** Visual support (*See Appendix A*), Labels for each round (Unfair, Random, Work, Proof of Work)

Equipment

- Timer or stopwatch
- Central table or container to act as the “bank”
- Open space for movement-based games

PROCEDURE

OPENING

1. Clear a suitably sized area for physical challenges.
2. Prepare a central “bank” with all beads.
3. Set up space for movement and ensure safety.
4. Prepare labels for each game round.
5. Lay out materials so transitions between rounds are quick.

PRE-ACTIVITY

5 minutes

1. Welcome students and set the tone with enthusiasm.
2. **Explain**: “Today we are going to play games, but not all of them will be fair.”
3. Introduce the idea of **My Stack**: students try to earn beads, but only keep beads from games the class agrees are fair.
4. **Ask** students: “What does fair mean to you?” “Can you think of a time when something felt unfair?” “How did it make you feel?”
5. **Set expectations** clearly: “We will play four short games. After each one, you will decide if it was fair.”

Objective In Student’s Words

- “Test out which games are fair and which ones aren’t, and why.”

ACTIVITY

45 minutes

Setup

1. Explain that the goal is to grow ‘My Stack’ by earning beads.
2. Explain that after each round the class will vote on whether the game was fair.
3. Explain that beads from unfair games must be returned to the bank.
4. Explain that once a fair game is identified, it may be repeated so students can keep their rewards.

Model

1. Demonstrate how beads are earned and returned to the bank.
2. Model respectful reactions to winning, losing, and disagreement.

Round 1: Unfair

10 minutes

1. Choose one unfair method such as a drawing competition judged by teacher preference, a biased game based on arbitrary traits, or unequal distribution based on attributes like height or name order.
2. Award beads according to the chosen method.
3. Ask: “Was that fair?” “Did everyone have the same chance?” “Did effort matter?”
4. Return all beads to the bank.

Round 2: Random

10 minutes

1. Choose a random reward method such as dice rolling or card drawing.
2. Distribute beads based on luck.
3. Ask: “Was this fair?” “Did effort decide the winner?”
4. If the class agrees it was unfair, return all beads to the bank.

Round 3: Work

10 minutes

1. Present physical or mental challenges such as jumping jacks, push-ups, staring contests, or arm wrestling.
2. Award beads for completing or winning tasks.
3. Ask: “Was this fair for everyone?” “Could everyone participate equally?”
4. Return beads unless the class agrees the system was fair.

Round 4: Proof of Work

10 minutes

1. Give each student five dice.
2. Set a clear target outcome such as rolling three matching numbers.
3. Students roll repeatedly until they succeed, re-rolling each time they fail.
4. Ask: “Can anyone win this game?” “Did everyone have the same chance?” “Does this feel fairer than the others?”
5. If agreed fair, allow students to keep beads and add them to **My Stack**.

Optional: assign a rotating student to verify results before rewards are confirmed.

Checkpoint

1. Pause briefly after each round to collect student judgments.
2. Record which systems were rejected and which were accepted as fair.

Reflection

- Ask students how each game made them feel.
- Ask what made the final game feel different.
- Reinforce that fairness is about rules, opportunity, and verification.

FOLLOW-UP

10 minutes

1. Review the four systems experienced: unfair, random, work-based, Proof of Work.
2. Discuss why Proof of Work allowed everyone the same chance.
3. Connect the experience to real-world systems where rewards are earned.

Optional story or creative reflection: students draw or write about a time something felt unfair or a system they think is fair.

CLOSE

5 minutes

1. Collect beads and materials.
2. Clear the activity space.
3. Thank students for engaging respectfully.
4. Final message to students: “Fair systems give everyone the same rules and the same chance.”

NOTES

Classroom management

- Acknowledge emotions openly and normalize frustration.
- Maintain calm control of bead distribution.
- Rotate roles to ensure inclusivity.

Extensions & Sponge Activities

- Retell the activity as a story or fable.
- Compare fairness in games vs fairness in real life.

Differentiation

- Younger students: fewer dice or shared rolls.
- ELL/Accessibility: visual cues, modeling, buddy system.
- Safety: supervise physical challenges and allow alternatives.

Appendices

- **Appendix A - Visual Aid**
- **Appendix B - Example unfair games**
- **Appendix C - Examples of Random Games**

2.4 That's Not Fair (Story Time)

Activity type Story time – Guided discussion – Moral reasoning

Duration 30–40 minutes

Grouping Whole Class

Description

Students listen to a teacher-read story in which characters receive rewards that may or may not match their effort. At key moments, the teacher pauses the story to invite discussion and judgment. Learners reflect on fairness, effort, luck, and unequal outcomes, building moral reasoning and vocabulary that supports later lessons on work, reward, and Proof of Work.

Learning Outcomes

By the end of this activity, learners will:

- Identify situations that feel fair or unfair
- Connect effort to reward through story events
- Practice expressing opinions and emotions respectfully
- Develop early moral reasoning using real examples

Materials

- One storybook selected by the teacher that includes effort and reward
- Drawing paper and pencils or crayons
- **Nice-to-have:** printed images of story characters, sticky notes to mark pause points in the story

Equipment

- Comfortable reading area (rug, chairs, circle seating)
- **Nice-to-have**: Visual timer to pace discussion pauses

PROCEDURE

OPENING

1. Choose a story that clearly includes characters who work hard, characters who do not, and rewards that can be questioned (See Appendix A & B).
2. Pre-read the story and mark 3–5 clear pause points where fairness can be discussed.
3. Arrange seating so all students can see and hear clearly.
4. Prepare drawing materials for the reflection activity.

PRE-ACTIVITY

2 minutes

1. Tell students: “Today I’m going to read you a story, but we are going to stop and think while we read.”
2. Ask: “What does fair mean?”
3. Ask: “Can something feel unfair even if it follows the rules?”
4. Explain the key question for the lesson: “As we read, we will keep asking one question: *Is that fair?*”

Objective In Student’s Words

- “Listen to a story and decide if the characters get what they deserve.”

ACTIVITY

20-25 minutes

Setup

1. Show the book cover and introduce the characters briefly.
2. Tell students that you will pause during the story to think together.

Model

1. Before reading, model a fairness judgment using a simple example: “If someone works hard and gets nothing, that might feel unfair.”
2. Remind students there are no right or wrong answers, only reasons.

Carry Out

1. Begin reading the story aloud.
2. At the first pause point, stop and ask: “What just happened?”
3. Ask: “Did this character work for that reward?”
4. Ask: “Does this feel fair? Why or why not?”
5. Continue reading and repeat this process at each marked pause point.
6. Encourage students to support their answers with reasons, not just opinions.

Checkpoint

1. Halfway through the story, pause and ask: “So far, who do you think is being treated fairly?”
2. Ask: “Who is not?”
3. Ask: “Do you think this will change by the end?”

Reflection

- After finishing the story, ask: “Which part felt the most unfair?”
- Ask: “Which character worked the hardest?”
- Ask: “Did effort always lead to reward?”

FOLLOW-UP

10 minutes

1. Review the main ideas: effort, reward, fairness.
2. Ask: “What makes a reward fair?”
3. Ask: “Should everyone get the same reward, or should effort matter?”
4. Connect forward: “In future lessons, we’ll explore systems where effort is what decides rewards.”

CLOSE

1. Collect drawing materials and tidy the reading space.
2. Thank students for listening and sharing their ideas.
3. Final wrap-up message to students: “Fairness is not just about outcomes, it’s about effort and opportunity.”

NOTES

Classroom management

- Normalize emotional reactions and disagreement
- Ensure all voices are heard
- Keep pauses short and focused

Extensions & Sponge Activities

- Students draw one scene and show whether it was fair or unfair.
- Rewrite the ending to make it fairer
- Act out one scene and vote on fairness

Differentiation

- Younger students: yes/no fairness voting with thumbs
- ELL/Accessibility: visual supports and simplified questions
- Safety: seated activity, minimal movement

Appendices

- **Appendix A - Example of a suitable story**
- **Appendix B - Other Examples**

2.5 Time Preference

Activity type Storytelling - Guided discussion - Comparison activity

Duration 20 minutes

Grouping Teacher-led, whole class

Description

Students explore the concept of time preference through a familiar story, *The Three Little Pigs*. By listening to the story, discussing character choices, and comparing building materials, learners reflect on short-term convenience versus long-term strength. The activity helps children understand that effort and patience can lead to better outcomes later.

Learning Outcomes

By the end of this activity, learners will:

- Recognize that choices made quickly can have consequences later
- Understand that effort and patience can lead to stronger outcomes
- Begin to grasp the idea of time preference in simple terms
- Practice comparing options and explaining their reasoning

Materials

- Storybook or printed visual version of *The Three Little Pigs*
- Visual aids showing houses made of straw, sticks, and bricks (*See Appendix A*)
- Crayons or markers
- **Nice-to-have:** Puppets or character cut-outs

Equipment

- Whiteboard or poster board for group discussion notes
- Markers
- **Nice-to-have**: Projector or screen for displaying images

PROCEDURE

OPENING

1. Prepare the storybook and visual aids so they are easy to show while reading.
2. Set up the board or poster space for writing student ideas.
3. Place handouts and crayons where they can be distributed quickly.

PRE-ACTIVITY

3 minutes

1. Ask: “Who knows the story of *The Three Little Pigs*?”
2. Ask: “What do you remember about the houses they built?”
3. Explain that the class will listen to a story and think about the choices each pig made.

Objective In Student’s Words

- “Listen to the story and decide which choices were smart and why.”

ACTIVITY

15 minutes

Setup

1. Explain that students will listen to a story, look at pictures, and talk about what was smart or not-so-smart.
2. Remind students to listen respectfully and take turns speaking.

Model

1. Before reading, give a simple example: “Sometimes the easy way feels good now, but the strong way helps later.”
2. Explain that the story will help them see the difference.

Carry Out

1. Read the Story
    1. Read *The Three Little Pigs* aloud using expressive voices.
    2. Show visuals for each house as it appears.
    3. Pause occasionally to ask: “What do you think will happen next?”
    4. Emphasize how quickly each house is built and how it stands up to the wolf.
2. Group Discussion
    1. Ask: “Which house was the strongest?”
    2. Ask: “Which house was built the fastest?”
    3. Ask: “Which pig spent the most time building?”
    4. Ask: “What happened when the wolf came?”
    5. Introduce choices and consequences by asking: “Was it a good idea to take the easy way?” and “Why did the brick house last?”
    6. Introduce the term **time preference** in simple language: “Time preference means choosing something easy now or something better later.”

3. Comparison Activity
    1. Give each student a comparison handout or display the chart on the board (*See Appendix A*).
    2. Review each material together: straw, sticks, bricks.
    3. Ask students to check the boxes they think fit each material.
    4. Ask: “Which house would you build?” and “Why?”

Reflection

- Ask students to explain which material lasts the longest.
- Ask students to explain which material is quickest to build.
- Listen for reasoning that connects effort to outcome.
- Ask: “Can you think of a time when waiting helped you?”

FOLLOW-UP

3-5 minutes

- Review the main idea: short-term choices can cause long-term problems.
- Apply time preference to everyday decisions such as saving treats or finishing work before playing.
- Reinforce that patience and effort can lead to better results.

CLOSE

2 minutes

1. Collect handouts and markers.
2. Clear the discussion space.
3. Thank students for participating.
4. Final message: “Working harder now can make life safer and easier later.”

NOTES

If teachers prefer, other stories can be selected for reading. The focus of this activity is to discuss the importance of how time preference can affect outcomes.

Classroom management

- Keep storytelling interactive with predictions and gestures
- Use pauses to manage attention and engagement
-

Extensions & Sponge Activities

- Use pauses to manage attention and engagement
- Discuss materials used today such as plastic, wood, leather, or stone

Differentiation

- Younger students: pointing, visuals, yes/no questions
- ELL/Accessibility: repetition, images, simplified language
- Safety: seated activity, minimal movement

REFERENCES

- *The Three Little Pigs* (traditional folktale)
  - https://en.wikipedia.org/wiki/The_Three_Little_Pigs

Appendices

- **Appendix A - Visual aids**
- **Appendix B - House comparison chart**

2.6 Smart Choices

Activity type Discussion - Visual demonstration - Creative craft - Simple game

Duration 60 minutes

Grouping Whole class for discussion, individual for craft, pairs or small groups for game and reflection

Description

Students explore how everyday choices affect long-term outcomes by connecting time preference to food, health, work, play, and education. Through discussion, a visual time-preference axis, creative superhero design, and a lighthearted comparison game, learners reflect on how small daily decisions build strength, focus, and future success.

Learning Outcomes

By the end of this activity, learners will:

- Understand that long-term thinking leads to stronger outcomes
- Apply time preference to everyday decisions
- Reflect on habits related to health, food, learning, and work
- Recognize that small choices compound over time

Materials

- Superhero templates (*See Appendix A*)
- Time Preference (*See Appendix B*)
- Colouring pencils, crayons, or markers
- Beads or tokens for rewards
- Candy or small treats (optional)

- Hero Stat scoring guide (*See Appendix C*)

Equipment

- Whiteboard or large poster space
- Markers
- Glue and scissors (optional)
- **Nice-to-have**: Timer or visual countdown clock

PROCEDURE

OPENING

1. Prepare the whiteboard with space for a time preference axis (*See Appendix B*).
2. Organize superhero templates, colouring tools, and game materials.
3. Prepare beads and optional treats.
4. Ensure clear space for group discussion and later game play.

PRE-ACTIVITY

10 minutes

1. Recap the previous lesson using *The Three Little Pigs*.
2. Ask: “Which house lasted the longest?” and “Why was it the strongest?”
3. Draw a horizontal line on the board to represent time preference.
4. Label the left side: High Time Preference - wants results now.
5. Label the right side: Low Time Preference - willing to wait for better results.Place straw, sticks, and bricks along the axis.

Objective In Student’s Words

- “Learn how my choices today can make me stronger later.”

ACTIVITY

45 minutes

Setup

1. Explain that everyday choices are like building materials.
2. Explain that some choices are fast and fun but weak, while others take effort but last longer.

Model

1. Give an example such as candy versus a healthy meal.
2. Place each example on the time preference axis and explain why.

Carry Out

1. Discussion: Everyday Choices

20 minutes

1. Introduce categories: food, health, work, play, education.
2. Give examples for each category such as candy versus vegetables or rushing homework versus doing it well.
3. Ask students where each choice belongs on the axis.
4. Invite students to justify their placements.

2. Design Your Superhero

25 minutes

1. Hand out superhero templates.
2. Instruct students to name their hero, colour the costume, and choose favourites in each category. Encourage creativity and imagination.
3. Review student choices and assign strength points to each hero based on time preference.

3. Superhero Battle and Reflection

15 minutes

1. Pair students or form small groups.
2. Have students compare hero stats in Top-Trumps style rounds.
3. Emphasize that the goal is learning, not winning.

Checkpoint

1. Observe whether students can explain why certain choices create stronger heroes.
2. Listen for connections between effort and long-term outcomes.

Reflection

- Ask: “Which choices made your hero strongest?”
- Ask: “Is there one choice you want to improve?”
- Invite students to name one low time preference action they could try this week.

FOLLOW-UP

10 minutes

1. Review the idea that small daily choices add up over time.
2. Reinforce that effort now often leads to better results later.
3. Connect to real life by asking students to identify one habit they can practice today.

CLOSE

5 minutes

1. Collect materials and clean up craft areas.
2. Thank students for their creativity and honesty.
3. Final message: “Strong futures are built one small choice at a time.”

NOTES

Classroom management

- Keep tone encouraging and non-judgmental
- Emphasize that habits are explored, not graded

Extensions & Sponge Activities

- Redesign the hero after one week of better choices
- Delay-of-gratification challenge using a small treat and a bonus reward for waiting

Differentiation

- Younger students: fewer choices and guided selection
- ELL/Accessibility: visuals, modeling, buddy support
- Safety: supervise scissors and movement

REFERENCES

- The Marshmallow Test: Mastering self-control.
  - https://psycnet.apa.org/record/2014-43233-000

Appendices

- **Appendix A - Superhero template**
- **Appendix B - Time preference axis examples**
- **Appendix C - Hero stat scoring guide**

Module 03

Analog vs. Digital

3.1 Back In My Day

3.2 Vote With Your Feet

3.3 Invent The Future

3.1 Back In My Day

Activity type Matching pairs – Coloring – Group discussion

Duration 20 minutes

Grouping Individual work with optional pair-share and whole class discussion

Description

Students match older analogue technologies with their modern digital equivalents. Through visual matching, colouring, and discussion, learners explore how tools evolve over time and how innovation changes the way people live, work, and communicate. The activity builds early awareness of technological progress and time-based change.

Learning Outcomes

By the end of this activity, learners will:

- Identify examples of analogue and digital technologies
- Understand that technology changes over time
- Recognize innovation as a process of improvement and adaptation
- Reflect on how technology affects daily life

Materials

- Printed matching worksheet (*See Appendix A*)
- Coloured pencils or crayons
- **Nice-to-have**: Visual Aids (*See Appendix B*)

Equipment

- Whiteboard or chalkboard
- Visual aid or slideshow
- **Nice-to-have**: Timer

PROCEDURE

OPENING

1. Ensure every student has a clear surface to work on.
2. Prepare the board or visual display with images of older technologies.
3. Distribute colouring materials in advance.
4. Introduce the activity by saying: “Today we’re going to look at how technology has changed from the past to today.”
5. Show examples such as a typewriter, Walkman, oil lamp, or VHS tape.

6. Ask: “Have you seen this before?” “What do we use instead today?”

PRE-ACTIVITY

2-3 minutes

1. Explain that students will act as technology detectives.
2. Introduce vocabulary: analogue, digital, technology, innovation, evolution.
3. Clarify meanings using simple explanations or images.
4. Explain that the goal is to match old tools with modern ones.

Objective In Student’s Words

- “Match old technology with new technology and see how things have changed.”

ACTIVITY

10 minutes

Setup

1. Distribute one matching worksheet to each student (*See Appendix A*)
2. Write simple instructions on the board.

Model

1. Demonstrate one example match on the board.
2. Explain why the two items belong together.

Carry Out

1. Students match each analogue item on the left with its digital version on the right.
2. Students colour the items they have used before.
3. Students circle items they have at home.
4. Teacher circulates to support students who do not recognize certain technologies.

Checkpoint

1. Observe whether students can correctly match most pairs.
2. Pause briefly if several students struggle and clarify with visuals.

FOLLOW-UP

3-5 minutes

1. Bring the class together for discussion.
2. Ask: “Which old technology did you find most interesting?”
3. Ask: “Which new technology do you use the most?”

4. Ask: “Are newer technologies always better?”
5. Optional pair-share: students tell a neighbour one surprising match they learned.

CLOSE

1. Collect worksheets and materials.
2. Thank students for their focus and effort.
3. Final message: “Technology changes over time, but every tool helps people solve problems in their own time.”

NOTES

Classroom management

- Prepare materials in advance to avoid interruptions
- Encourage early finishers to colour carefully and reflect

Extensions & Sponge Activities

- Home interview with parents or grandparents about old technology
- Invent and draw a future version of one tool
- Watch short demonstration videos of older technologies

Differentiation

- Younger students: guided matching with board visuals
- ELL/Accessibility: images, simplified language, peer support
- Safety: seated activity, minimal movement

Appendices

- **Appendix A - Analog to digital matching pairs**
- **Appendix B - Visual Aids**
- **Appendix C - Have you Ever used...?**

3.2 Vote With Your Feet

Activity type Debate - Movement-based decision making

Duration 20-25 minutes

Grouping Whole class, teacher-led, optional team split for extension

Description

Students physically position themselves in the classroom to show whether they prefer older or newer technologies. By moving, debating, and explaining their choices, learners compare analogue and digital

tools, consider trade-offs, and practice respectful disagreement. The activity reinforces prior learning on technological evolution while increasing engagement through movement.

Learning Outcomes

By the end of this activity, learners will:

- Compare old and new technologies and explain preferences
- Identify advantages and disadvantages of different tools
- Practice expressing opinions and listening to others
- Engage in cooperative discussion through movement and debate

Materials

- Completed Analogue vs Digital matching worksheets (*See 3.1 - Evolution of Technology*)
- Optional labels or signs for OLD, CENTER, NEW
- Optional beads or stars for debate rewards
- Optional poster paper and markers

Equipment

- Clear classroom space for movement
- **Nice-to-have**: Timer or bell for transitions

PROCEDURE

OPENING

1. Confirm students have completed the Analogue vs Digital matching worksheet.
2. Ask students to retrieve and briefly review their matches.
3. Explain the purpose of the activity: “Now we’re going to share our opinions about old and new technologies.”
4. Emphasize that there are no right or wrong answers.

PRE-ACTIVITY

3-5 minutes

1. Lead a short whole-class discussion about whether new technologies are always better.
2. Ask what old technologies are good at and what might be lost when upgrading.
3. Select one technology pair such as typewriter versus laptop or horse and cart versus car.
4. Invite students to name pros and cons of each option.
5. Take a quick show of hands for initial preference.

Objective In Student's Words

- "Show which technology I prefer and explain why."

ACTIVITY

10-15 minutes

Setup

1. Designate three physical zones in the classroom labeled OLD, CENTER, and NEW.
2. Ensure boundaries are clear and movement rules are understood.

Model

1. Demonstrate the process using an easy example such as letter versus email.
2. Model moving to a side and giving a short explanation.
3. Reinforce respectful listening and disagreement.

Carry Out

1. Read aloud one pair of technologies.
2. Count down clearly and say "GO."
3. Students move to OLD, NEW, or CENTER based on preference.
4. Invite one or two students from each side to explain their choice.
5. Encourage respectful disagreement and reasoning.
6. Ask all students to return to the CENTER.
7. Repeat with new technology pairs for four to six rounds or while engagement remains high.

Checkpoint

1. Observe whether students can give reasons beyond simple preference statements.
2. Pause to clarify misunderstandings about technologies when needed.

Reflection

- Ask whether anyone changed their mind after hearing from others.
- Ask which technology pair was hardest to choose.
- Reinforce that different preferences can still be valid.

FOLLOW-UP

3-5 minutes

- Review that technology changes over time and serves different needs.
- Emphasize that both old and new tools have strengths.
- Connect the discussion to future lessons on innovation and adaptation.

Optional extensions include short debates, assigned arguments, or creating a class vote chart.

CLOSE

2 minutes

1. Ask students to return to their seats.
2. Remove signs and reset the classroom layout.
3. Thank students for respectful participation.
4. Final message to students: “Every technology has its time and place, and what matters is how we think about and use our tools.”

NOTES

- Use this lesson to foster critical thinking *without judgment*
- Balance fun and focus - encourage physical activity while maintaining classroom order
- Consider using real or printed images of the items to support visual learners
- This activity can also segue into a future lesson about *innovation*, *change over time*, or *how people adapt to new tools*

Classroom management

- Enforce safe movement and clear boundaries
- Watch for quieter students and invite them to speak

Extensions & Sponge Activities

- Visual bar graph of class preferences
- Journal reflection on favorite technology
- Instead of choosing, assign students a technology to argue *for*
- Create a class bar graph or poster

Differentiation

- Younger students: fewer rounds and simpler explanations.
- Older students: Invite 1–2 volunteers from each side to “defend” their choice
- ELL/Accessibility: visual cues, modeling, and sentence starters.
- Safety: Safety: no running and clear movement rules.

3.3 Invent The Future

Activity type Creative invention - Pitch presentation - Structured discussion

Duration 45-60 minutes

Grouping Individual or small teams of 2–4 students

Description

Students imagine the future of technology by inventing a new product or system and pitching it in a "Dragons' Den" format. Learners reflect on how technology evolves, identify real problems to solve, and practice explaining ideas clearly. The activity encourages creativity, critical thinking, and communication while reinforcing the idea that innovation responds to needs over time.

Learning Outcomes

By the end of this activity, learners will:

- Predict how technology might evolve in the future
- Design a simple invention to solve a problem
- Explain how and why their invention works
- Practice presenting and defending an idea.

Materials

- Blank paper for sketches and notes
- Pencils, crayons, or markers
- **Nice-to-have:** Pitch Template (*See Appendix A*)

Equipment

- Whiteboard or poster space
- **Nice-to-have**: Timer for pitches

PROCEDURE

OPENING

1. Arrange seating so teams can work together and later present to the front.
2. Prepare board space to list examples of past, present, and emerging technologies.
3. Decide how pitches will be timed and assessed.

PRE-ACTIVITY

5-10 minutes

1. Show or describe examples of inventions that succeeded and inventions that failed.
2. Discuss briefly why some technologies worked and others did not.
3. Introduce a few emerging technologies and ask students what they think might happen next.
4. Explain that today students will become inventors predicting the future.

Objective In Student's Words

- "Invent a new technology and convince others that it's a good idea."

ACTIVITY

30-40 minutes

Setup

1. Explain the Dragons' Den format: students invent, then pitch.
2. Explain expectations for the pitch: what it is, what problem it solves, and why it matters.
3. Assign students to work individually or in small teams.

Model

1. Give a simple example invention such as a backpack that charges devices using movement.
2. Model a short pitch explaining the problem, solution, and benefit.

Carry Out

1. Students brainstorm problems they want to solve.
2. Students design an invention that addresses the problem.
3. Students sketch the invention and write key points for their pitch.
4. Teams rehearse how they will explain their idea.
5. Students present their invention to the teacher or class in short pitches.

Checkpoint

1. Circulate during planning to ensure ideas are clear and realistic.
2. Ask guiding questions such as "Who would use this?" and "Why is this better than what exists now?"

Reflection

- After presentations, ask students what inventions they found most interesting.
- Ask what problems appeared most important to solve.
- Ask what made a pitch convincing.

FOLLOW-UP

5-10 minutes

1. Review that technology evolves to solve problems.
2. Discuss how predictions can be right or wrong over time.
3. Connect to real-world innovation and how ideas improve through feedback.

CLOSE

1. Collect sketches and materials.
2. Reset the classroom.

3. Thank students for their creativity and effort.
4. Final message: “Every invention starts as an idea someone was brave enough to share.”

NOTES

Classroom management

- Keep pitch timing consistent.
- Encourage respectful listening and questions.

Extensions & Sponge Activities

- Turn the pitch into a written assignment.
- Add voting categories such as most useful, most creative, or most realistic.

Differentiation

- Younger students: focus on drawing and simple explanations.
- ELL/Accessibility: visual support, partner pitching.
- Safety: seated activity, no physical construction required.

Appendices

- **Appendix A - Pitch template**

Module 04

Safety, Security & Privacy

4.1 Gatekeepers

4.2 Wire Tap

4.3 Cypherpunks Write Code

4.4 Nostr

4.1 Gatekeepers

Activity type Guided discussion - Visual classification - Personal reflection

Duration 30 minutes

Grouping Whole Class/Pairs/Individual

Description

Students explore the difference between public and private spaces and information through guided discussion and visual examples. Using familiar everyday situations, learners practice classifying what is usually public or private and reflect on the idea that individuals can choose what information they keep private or share. The activity builds early awareness of privacy, boundaries, and personal responsibility.

Learning Outcomes

By the end of this activity, learners will:

- Distinguish between public and private spaces and information
- Explain why some things should be kept private
- Recognize that people can choose what they share
- Practice respectful discussion and personal decision-making

Materials

- Visual aids or flash cards showing examples of places and information (*See Appendix A*)
- Student worksheet with checkboxes for Public / Private (*See Appendix B*)
- Pencils or crayons

Equipment

- Whiteboard or poster board
- Markers
- **Nice-to-have:** Projector or screen for images

PROCEDURE

OPENING

1. Prepare visual aids or slides showing examples such as a park, bus, washroom, car, eye colour, password, bank PIN.
2. Prepare the board with two headings: PUBLIC and PRIVATE.
3. Place worksheets and writing materials where they can be easily distributed.

PRE-ACTIVITY

5 minutes

1. Ask students: “What does ‘public’ mean?” and “What does ‘private’ mean?”
2. Accept multiple answers and do not correct yet.
3. Explain that today they will look at everyday examples and decide together.

Objective In Student’s Words

- “Assess what things are public, what things are private, and what I want to share.”

ACTIVITY

20 minutes

Setup

1. Explain that some things are usually public, some are usually private, and some depend on choice.
2. Introduce the idea that there is not always one correct answer.

Model

1. Show one example such as a park.
2. Think aloud: “Many people can go there and see it, so we usually say this is public.”
3. Show a second example such as a password.
4. Think aloud: “This belongs only to one person, so we usually keep it private.”

Carry Out

1. Show one visual example at a time.
2. Ask students to vote or raise their hands for ‘public’ or ‘private’.
3. Place the example under the matching heading on the board.
4. Ask follow-up questions such as “Why?” or “Could this ever change?”
5. Use examples such as park, bus, washroom, car, eye colour, password, bank PIN.

Checkpoint

1. Ask students to explain the difference between public and private in their own words.
2. Listen for understanding rather than memorized definitions.

Reflection

- Ask: “Is eye colour always private?”
- Ask: “Can something be private even if other people can see it?”
- Guide students toward the idea that people choose what information they share.

FOLLOW-UP

5 minutes

1. Distribute the checkbox worksheet.
2. Students decide for each item whether they would keep it public or private.
3. Emphasize that answers may be different for different people.

CLOSE

1. Collect worksheets and materials.
2. Summarize key idea: “You get to decide what information you share.”
3. Final message to students: “Being smart about privacy helps keep you safe.”

NOTES

Classroom management

- Encourage respectful listening and differing opinions
- Avoid asking students to share sensitive personal information

Extensions & Sponge Activities

- Add digital examples such as usernames or photos
- Create a class rule poster for sharing information
- Look at super-heroes and villains - why do they protect their identity?
- Make a diagram of information each students wants to share/keep private

Differentiation

- Younger students: use only place-based examples
- Older students: move to more difficult subjects (DNA, Facial Recognition, GPS location, Name)
- ELL/Accessibility: visuals, gestures, simple language
- Safety: avoid asking for real passwords or personal data

Appendices

- **Appendix A - Visual Aid**
- **Appendix B - Public vs Private worksheet**

4.2 Wire Tap

Activity type Simulation game - Guided discussion - Experiential learning

Duration 40-50 minutes

Grouping Small teams, whole class discussion

Description

Students explore how information changes depending on who carries it and who is allowed to hear it. Through a series of short message-passing games with changing rules, learners experience trust, mistakes, and intentional interference firsthand. Reflection focuses on how messages can change, leak, or be protected, and who should decide who gets access to information.

Learning Outcomes

By the end of this activity, learners will:

- Experience how information can change when passed between people
- Recognize the difference between mistakes and intentional changes
- Understand that not everyone should hear every message
- Begin thinking critically about trust, privacy, and control of information

Materials

- Prepared short and long message cards (for teacher reference) (*See Appendix A*)
- Paper and pencil for running dictation variant
- **Nice-to-have:** Simple symbol or substitution code sheet for optional locked message (*See Appendix B*)

Equipment

- Open classroom space for movement
- Whiteboard or poster board for reflection notes
- **Nice-to-have**: Timer

PROCEDURE

OPENING

1. Prepare simple and complex messages in advance.
2. Decide team groupings of 6–8 students.
3. Clear space so students can line up or move safely.

PRE-ACTIVITY

5 minutes

1. Ask students if they have ever played telephone.
2. Ask whether the message always stays the same.
3. Explain that today they will test who can be trusted with a message.
4. Avoid introducing technical or abstract terms.

Objective In Student's Words

- "See what happens to a message when different people carry it."

ACTIVITY

30-35 minutes

Setup

1. Divide the class into small teams.
2. Explain that each team will run several short rounds with the same structure but different rules.
3. Emphasize that no one is in trouble for mistakes.

Model

1. Demonstrate whispering rules using a very short example.
2. Reinforce rules clearly: whisper only, no repeating, no fixing mistakes.

Carry Out

Round 1 – Easy Message (Trusted Chain)

1. Whisper a very simple sentence to the first student.
2. Students pass the message by whispering down the line.
3. The last student says the message out loud.
4. Checkpoint: Was the message correct, did anyone change it on purpose, did it feel easy to trust the group?

Round 2 – Hard Message (Complex Information)

1. Use the same setup with a longer, more precise sentence.
2. Run the whisper chain again.
3. The last student says the message out loud.
4. Checkpoint: What changed, where did it break, was anyone being bad or was it just hard?

Round 3 – Hidden Rule (Intentional Change)

1. Use the same complex message.
2. Secretly tell one student in the middle they may change one word.
3. Do not announce that this rule exists.
4. Run the whisper chain.
5. Checkpoint: Did this feel different from mistakes, how could we tell something was wrong, what felt unfair?

Optional Variant – Running Dictation (Control vs Trust)

Round 4 – Trusted Runner

1. Place the message written on paper across the room.
2. One student runs to read it and returns to tell the writer.
3. The writer records the message.
4. Checkpoint: Was this clearer than whispering and why?

Round 5 – Untrusted Space

1. Add a listener near the runner's path.
2. The listener may hear but must not see the message.
3. Checkpoint: Who learned something they shouldn't have and was the message still safe?

Optional Final Twist – Locked Message

1. Repeat running dictation with the message written in a simple symbol code (*See Appendix B*).
2. Ask students to observe who can understand it.
3. Checkpoint: Who can read it and who hears it but learns nothing?

Reflection

1. Ask who should hear everything.
2. Ask who should hear nothing.
3. Ask who should decide.
4. Allow students to name the problem in their own words.

FOLLOW-UP

5-10 minutes

1. Review that messages can change, leak, or be altered.
2. Reinforce that trust depends on rules and boundaries.
3. Connect to earlier lessons on public versus private information.

CLOSE

5 minutes

1. Thank students for participating honestly.
2. Reset the classroom layout.
3. Final message to students: Being careful with information helps protect people.

NOTES

Classroom management

- Normalize mistakes and emotional reactions while maintaining calm control.

- Do not accuse or reveal the hidden rule-breaker unless students explicitly ask.

Extensions & Sponge Activities

- Write a class rule for protecting messages or redesign the game to make it fairer.

Differentiation

- Younger students: shorter messages and fewer rounds.
- ELL/Accessibility: visual cues, slower pacing, buddy support.
- Safety: controlled movement and clear whispering rules.

Appendices

- **Appendix A - Example simple and complex messages**
- **Appendix B - Simple symbol code**

4.3 Cypherpunks Write Code

Activity type Craft - Individual decoding activity - Guided practice

Duration 45-60 minutes

Grouping Individual work with whole class demonstration

Description

Students are introduced to the idea of codes as a way to hide and protect information. Through a hands-on craft, learners build a simple cipher wheel and use it to encode and decode Bitcoin-related vocabulary. The activity combines fine motor skills, logical thinking, and early cryptographic concepts while reinforcing new terminology in a playful and concrete way.

Learning Outcomes

By the end of this activity, learners will:

- Understand the meaning of encode and decode; recognize that changing a code changes the output; use a cipher wheel to encode and decode simple words; correctly decode and encode selected Bitcoin vocabulary.

Materials

- Printed cipher wheel templates with inner and outer alphabet rings (*See Appendix A*)
- Printed worksheet with encoded Bitcoin vocabulary (*See Appendix B*)
- Pencils, crayons, or markers
- **Nice-to-have:** Pre-made cipher wheel for demo, Brads or split pins for fastening wheels, Extra practice worksheets

Equipment

- Scissors
- Whiteboard or poster board
- **Nice-to-have**: Projector demonstration, Left-handed scissors

PROCEDURE

OPENING

Teacher setup

1. Print and prepare cipher wheel templates and worksheets.
2. Pre-cut one large demonstration cipher wheel or prepare a visual version for the board.
3. Ensure scissors and pins are safely distributed.

PRE-ACTIVITY

5 minutes

1. Ask students if they know what a secret code is.
2. Ask where they have seen codes before such as games, secret messages, or locks.
3. Introduce the words encode and decode using simple language.
4. Explain that today they will learn how to hide and reveal words using a code.

Objective In Student's Words

- "Make a secret code tool and use it to hide and read words."

ACTIVITY

35-45 minutes

Setup

1. Explain that letters can be shifted to make secret messages.
2. Draw the alphabet on the board and show a simple letter shift such as A becomes D.

Model

1. Demonstrate encoding a short word using a letter shift.
2. Demonstrate decoding the same word back to the original.
3. Introduce the cipher wheel and show how turning it changes the output letters.
4. Explain that the number of steps the wheel is turned changes the result.

Carry Out

Part 1 – Cipher Wheel Craft

1. Students cut out the inner and outer alphabet circles.
2. Students place the smaller circle on top of the larger one.
3. Students fasten the circles together with a pin so the wheel can turn freely.
4. Students test the wheel by aligning different letters.

Checkpoint

1. Ask students to decode one word together as a class.
2. Check that students understand how changing the wheel changes the result.

Part 2 – Encode and Decode Practice

1. Students use their cipher wheel to decode words on the worksheet.
2. Students encode given Bitcoin vocabulary words using a chosen letter shift.
3. Circulates to support students and check understanding.

Reflection

- Ask students what happened when they changed the wheel position.
- Ask whether the same word always looks the same when the code changes.
- Ask why someone might want to hide information this way.

FOLLOW-UP

2 minutes

- Review key terms: encode, decode, code, message.

CLOSE

- Collect scissors and materials.
- Allow students to keep their cipher wheels.
- Final message to students: “Changing the code changes the message, but the meaning stays the same.”

NOTES

Classroom management

- Supervise scissor and pin use carefully.
- Encourage patience during the craft stage.

Extensions & Sponge Activities

- Timed decoding challenge using new encoded words.

- Play a decoding challenge such as cipher hangman where the encoded word is written on the board.

Differentiation

- Younger students: Shorter words and fixed letter shift, Use non-Bitcoin vocabulary
- Older students: Choose and record their own letter shift for others to solve.
- ELL/Accessibility: visual alphabet guides and peer support.
- Safety: blunt scissors and monitored pin use.

Appendices

- **Appendix A - Cipher wheel template**
- **Appendix B - Bitcoin vocabulary worksheet**
- **Appendix C - 40–60 Bitcoin vocabulary word cards**

4.4 Nostr

Activity type Simulation game – Craft-based encoding – Competitive challenge

Duration 50-60 minutes

Grouping Small teams of three students

Description

Building on earlier activities about trust, message passing, and interception, students now compare open information transfer with encoded communication. Learners first experience how relaying information requires trusting intermediaries and how messages can be overheard or altered. They then introduce cipher wheels to protect messages, discovering that information can remain private even when passed through untrusted or public channels.

Pre-requisites

- 4.2 - Telephone
- 4.3 - Cypherpunks Write Code

Learning Outcomes

By the end of this activity, learners will:

- Review how information transmission requires trust in intermediaries
- Recognize how messages can leak or change when transmitted openly;
- Understand that encoding protects information even when intermediaries are untrusted
- Successfully encode and decode vocabulary using a cipher wheel.

Materials

- Prepared word or sentence cards for transmission
- Cipher wheel templates for each student (*See Appendix A*)
- Worksheets or blank paper for transcription
- Pencils or markers
- **Nice-to-have:** Example simple and complex messages (*See Appendix B*)

Equipment

- Scissors
- Split pins or brads for cipher wheels
- Open classroom space for movement
- **Nice-to-have**: Timer

PROCEDURE

OPENING

1. Arrange the room with two tables per team placed far apart to create distance.
2. Place open message cards at each sender table.
3. Prepare cipher wheel materials but keep them hidden until Phase 2.

PRE-ACTIVITY

5 minutes

1. Ask students to recall the previous activity where messages were whispered or relayed.
2. Ask what problems they noticed with trust, mistakes, or overhearing.
3. Remind students that last time they could not keep messages private.
4. Explain that today they will test whether that problem can be solved.

Objective In Student's Words

- "Find a better way to send messages without everyone needing to be trusted."

ACTIVITY

40–45 minutes

Setup

1. Divide students into teams of three.
2. Assign roles: Player 1 is the Sender, Player 2 is the Runner, Player 3 is the Receiver.
3. Explain that roles will stay the same throughout the activity.

Model

1. Demonstrate a short running dictation round using an open message.
2. Point out that the runner must remember and repeat the message exactly.
3. Remind students that anyone nearby can overhear the message.

Phase 1 – Open Transmission (Trust Required)

1. Player 1 silently reads the message at the sender table.
2. Player 1 quietly reads the message to Player 2 at the sender table.
3. Player 2 runs to Player 3 and relays the message verbally.
4. Player 3 writes down what they hear.
5. Player 2 may run back and forth as needed.
6. The first team to successfully transmit the message wins.

Checkpoint Discussion

- Ask whether the final message matched the original.
- Ask whether runners overheard other teams.
- Ask whether the runner had the power to change the message.
- Reinforce that the system only worked if the runner was trusted.

Phase 2 – Encoded Transmission (Trust Reduced)

1. Distribute cipher wheel materials and allow students to assemble them.
2. Sender and Receiver privately agree on a letter shift using the wheel.
3. Player 1 encodes a new message using the agreed cipher.
4. Player 2 relays only the encoded message without knowing its meaning.
5. Player 3 transcribes and decodes the message using the cipher wheel.

Competition Rules

- Teams earn points for correct decoding.
- Speed and accuracy both matter.
- Runners are not allowed to learn the cipher.

Reflection

1. Ask what changed between open and encoded transmission.
2. Ask whether trust in the runner was still necessary.
3. Ask who controlled access to the message in each phase.

FOLLOW-UP

5-10 minutes

1. Review that earlier activities showed how messages can be intercepted or altered.
2. Review that encoding allows messages to travel through public space safely.
3. Connect this to the idea of choosing what information is public or private.

CLOSE

5 minutes

1. Collect shared materials and reset the room.
2. Allow students to keep their cipher wheels.
3. Final message to students: “You don’t need to trust everyone when your message is protected.”

NOTES

Classroom management

- Older Students can choose their role in the relay
- Control running paths and enforce walking if needed.
- Emphasize that mistakes are expected and part of learning.

Extensions & Sponge Activities

- Rotate roles and repeat with a new cipher.
- Increase message length or add decoy messages.

Differentiation

- Younger students: single words and fixed letter shift.
- ELL/Accessibility: alphabet charts and slower pacing.
- Safety: clear movement rules and supervised tool use.

Appendices

- **Appendix A - 40–60 Bitcoin vocabulary word cards**
- **Appendix B - Cipher wheel template**

Module 05

Money Matters

5.1 Money Matters

5.1 Money Matters

Activity type Simulation • LARP • Experiential Learning

Duration 90 minutes

Grouping Individual with cooperative learning

Description

Money Matters is a classroom economy simulation where each learner operates from a personal Home base. Each student begins with an individual HOME tile displaying their name, challenge card, and financial objective. Learners move through a shared economic system using dice-based movement rules to earn, save, invest, borrow, and trade. The economy is introduced progressively before opening into autonomous role-play, where learners pursue individual goals under shared rules, time constraints, and access limits. Optional extensions allow the system to scale into a richer economy with businesses, infrastructure costs, and education.

Learning Outcomes

By the end of this activity, learners will:

- Identify different ways people earn money
- Explain saving, investing, borrowing, and trade
- Understand how rules, chance, and access affect outcomes
- Track progress toward a visible financial objective
- Reflect on effort, risk, fairness, and decision-making

Materials

- Individual HOME tiles (paper or card stock, one per student)
- Sticky tack
- Pens or markers
- Gold coin tokens
- Dice
- Calculators
- Playing cards (52-card deck)
- Challenge index cards
- Grain and flour tokens/cards
- Bank rate card
- **Nice-to-have:** Timer or visible clock

Equipment

- Tables or desks arranged as stations

- Clear classroom space for movement
- **Nice-to-have**:

PROCEDURE

OPENING

PRE-ACTIVITY

45 minutes

Introduce the idea of a small economy where everyone starts from their own home. Explain that Home is a personal base for storage, planning, and tracking progress. Explain that movement, access, and opportunities are limited by rules and chance, and that everyone plays under the same system.

1. HOME TILE SETUP (10 minutes)

Each learner:

1. Receives one HOME tile
2. Writes their name clearly
3. Draws one card from the deck
4. Looks up their challenge
5. Attaches the card to the HOME tile using sticky tack
6. Checks their objective (SAVE, INVEST, or BORROW)
7. Writes the target amount on the HOME tile

Explain that the HOME tile stays in place, objectives remain visible, and learners update their progress honestly.

1. STATION INTRODUCTIONS (35 minutes)

Stations are introduced one by one:

1. Home
2. Farm
3. Factory
4. Market
5. Bank

After Town Square is introduced, demonstrate dice-based movement rules. Learners practice rolling to move from Home to Town Square and from Town Square to a destination. Emphasize that failed rolls cost time, not punishment.

Objective In Student's Words

“I start at my home and work toward my money goal.”

1.
2.

Objective In Student’s Words

- “I start at home and work toward my money goal.”

ACTIVITY

30 minutes

Setup

1. All stations are open.
2. HOME tiles are placed around the room or table layout.
3. Town Square acts as the central hub.
4. Learners begin at their own HOME tile.

Model

Teacher demonstrates:

1. Writing an objective on a HOME tile
2. Rolling to move
3. Completing one earning or trading action
4. Returning Home to record progress

Carry Out

1. Learners take turns rolling, moving, and completing actions. Only one movement roll is allowed per turn. If the roll fails, the learner remains in place. When learners return Home, they update the amount earned on their HOME tile. Challenge cards remain attached at all times.

Checkpoint

1. The teacher announces the remaining time halfway through the activity.

Reflection

- Learners consider how often they returned Home, which stations they used most, and how movement rules affected their choices.

FOLLOW-UP

15 minutes

Whole-class discussion:

- Who completed their challenge?
- Which objectives were hardest?
- How did saving, investing, or borrowing feel different?
- How did access and chance affect outcomes?

CLOSE

- Did having a visible goal change how you played?
- Did luck or planning matter more?
- What would you do differently next time?
- Are there any modifications to the economy you would like to see?

NOTES

Due to the complexity of this activity, this game can be spread across two sections Part 1 & Part 2. Also, once the rules have been understood, this game can be repeated over and over as many times as students are interested.

Classroom management

- HOME tiles remain fixed
- Learners update only their own HOME tile
- One active role per station where applicable
- Teacher intervenes only for rule enforcement

Extensions & Sponge Activities

- **Businesses**

Add new business stations such as Bakery, Mill, or Workshop. Businesses buy inputs (grain or flour), add value, and sell finished goods at the Market. Owners earn profit after costs, introducing entrepreneurship and margins.

- **Toll Roads** Add toll points between Town Square and key stations. Learners must pay 1–2 gold to pass or wait and attempt a different route. This models infrastructure costs, access friction, and time trade-offs.
- **Multiple Farm Businesses** Add two or more Farm stations with different rules. Examples include a fast farm with lower payout, a slow farm with higher payout, or a risky farm with higher variance. Learners choose based on risk and time.
- **Schools** Add a School station where learners can spend time or gold to gain temporary advantages such as an extra dice roll, faster factory work, or better bank rates. Benefits are limited in duration, highlighting long-term investment.

Differentiation

- Younger students: Smaller target amounts, Optional rerolls, Fewer active stations
- ELL/Accessibility: Visual icons on stations and HOME tiles, Peer support for reading challenges
- Safety: Walking only between stations, Clear movement paths maintained

Appendices

- **Appendix A - Student Card**
- **Appendix B - Challenge Cards (Legend)**
- **Appendix C - Movement Rules Table**

Module 06

Bitcoin Basics

6.1 From Problem to Protocol

6.2 Satoshi, Where Are You?

6.3 It's Mine!

6.1 From Problem to Protocol

Activity type Storytelling – Guided Discussion – Reflection

Duration 45-60 minutes

Grouping Whole Class/Pairs/Individual

Description

Students listen to the story of Satoshi and the Genesis of Bitcoin. Through guided pauses and discussion, they connect the narrative to prior lessons on value, scarcity, time and energy, analogue vs digital systems, and safety and privacy. The teacher uses a printed "white paper" prop to dramatize the moment Satoshi presents his idea to the world. Students reflect on how Bitcoin began as a proposal, required hard work to build, and depends on participants to keep the network alive.

Learning Outcomes

By the end of this activity, learners will:

- Explain who Satoshi is in simple terms.
- Describe what a white paper is.
- Identify the Genesis Block as the beginning of Bitcoin.
- Explain why Bitcoin requires time, energy, and shared rules.
- Recognize that networks stay alive when people participate.

Materials

- Printed copy of the story (*See Appendix A*)
- Printed "White Paper" prop (*See Appendix B*)
- Three keyword cards: VALUE - TIME & ENERGY - TRUST & PRIVACY (*See Appendix C*)
- Whiteboard and markers
- Exit slips or small paper
- **Nice-to-have:** Printed image of a chain of blocks, Small envelope labeled 'BTC'

Equipment

- **Nice-to-have**: Whiteboard or Projector (optional)

PROCEDURE

OPENING

- Print the story and place the "White Paper" prop inside an envelope labeled BTC for a dramatic reveal.
- Write on the board: What happens when money systems fail?

- Prepare three keyword cards: VALUE • TIME & ENERGY • TRUST & PRIVACY.
- Draw a simple "BANK – CLOSED" sign on the board to spark curiosity.
- Pre-select and mark 3–4 pause points in the story for guided discussion.

PRE-ACTIVITY

5 minutes

1. Ask students: "What would happen if a bank closed and people couldn't get their money?"
2. Write responses on the board. Connect briefly to previous chapters (Value, Scarcity, Trust)
3. Do not explain Bitcoin yet. Build curiosity.
4. Show students the sealed envelope labeled BTC.
5. Say: "Inside this envelope is an idea that changed money forever."

Objective In Student's Words

- "We are going to learn how Bitcoin started and why people choose to keep it alive."

ACTIVITY

20–25 minutes

Setup

1. Students sit in a circle or attentive listening formation.
2. Holds the printed story and white paper prop.

Carry Out

1. Read the story aloud with natural pacing.
2. Pause briefly at key moments:
    - When the bank closes – What is the problem?
    - When Satoshi works late – Why is effort important?
    - When the white paper is presented – What does it mean to share an idea with the world?
    - When Genesis Block is mentioned – What does beginning mean?
    - When the network must stay alive – Who keeps it alive?
3. Keep pauses short to preserve immersion.

Reflection

- Whole class discussion:
    - Why does Bitcoin need participants?
    - What happens if people stop caring about the rules?

- Connect to prior chapters:
  - Value – Why must Bitcoin be scarce?
  - Time & Energy – Why does Proof of Work matter?
  - Analogue vs Digital – How is Bitcoin different from paper money?
  - Safety & Privacy – Why must we protect keys?

FOLLOW-UP

2 minutes

- Students complete one exit sentence:
  - Bitcoin began when _____.
  - Satoshi presented _____ to the world.
  - The network stays alive when _____.
- Collect responses.
- Set-up the next activity by mentioning that we will explore where Satoshi disappeared to.

CLOSE

- Collect exit slips and any reflection sheets.
- Ask students to return printed story copies, prompt cards, and materials to the designated area.
- Have students neatly stack chairs or reset the seating layout if it was changed for storytelling.
- Ensure the white paper prop and appendix materials are stored safely for future use.
- Prepare the room for the next class or transition students calmly to the next scheduled activity.

NOTES

Classroom management

- Set clear listening expectations.
- Use call-back signals for transitions.
- Keep discussion focused and time-bound.

Extensions & Sponge Activities

- Write a letter to Satoshi explaining how you would protect the future of money.
- Draw the Genesis Block using three symbols: chain, clock, lock.
- Roleplay Satoshi presenting the white paper.

Differentiation

- Younger students: Shorten reading segments and focus on problem–idea–solution structure.
- ELL/Accessibility: Pre-teach vocabulary: proposal, genesis, network, effort, participant.
- Safety: Provide sentence starters for exit tasks.

Appendices

- **Appendix A - Mina and the Lemonade Stand (Example Story)**
- **Appendix B - Simplified White Paper Prop**
- **Appendix C - Keyword Cards**

6.2 Satoshi, Where Are You?

Activity type Creative Writing – Collaborative Learning – Dramatic Arts

Duration 60 minutes

Grouping Individual / Small Groups

Description

In this lesson, students construct a story using four independent ideas written on sticky notes. They begin by recalling the earlier story in which Satoshi disappeared and speculate about where he might have gone. Students then generate four ideas from guided prompts and organize them into a beginning, middle, and end structure. Through structured sharing and a sticky note swap, they revise their narratives to include new constraints. The lesson concludes with either a written or illustrated final draft, or a short dramatic role play using a class-generated story arc.

Learning Outcomes

By the end of this activity, learners will:

- Sequence ideas into beginning, middle, and end
- Construct a logical narrative from independent elements
- Adapt a story when new constraints are introduced
- Participate in structured peer storytelling
- Collaborate in a short dramatic performance

Materials

- Sticky notes (4 per student)
- Markers or pencils
- Handout with three columns: Beginning / Middle / End (*See Appendix A*)
- Whiteboard
- Timer
- **Nice-to-have:** Colored sticky notes, TImer

Equipment

- Whiteboard
- Markers

- **Nice-to-have**: Bell or signal for transitions

PROCEDURE

OPENING

- Prepare whiteboard space for later three-column setup.
- Ensure markers are accessible.
- Confirm timing signals and group arrangements.

PRE-ACTIVITY

8 minutes

1. Activate prior learning by referencing the earlier story where Satoshi disappeared.
2. Ask:
    - Where do you think Satoshi went?
    - Why might he have left?
    - What could have happened next?
3. Take brief responses.
4. Explain that today students will build their own stories using small ideas that connect together.
5. Ask one student to explain the task in their own words.

Objective In Student's Words

- "We are going to turn small ideas into a full story."

ACTIVITY

30 minutes

Carry Out

1. Distribute Materials - four sticky notes per student and markers. Explain the rule: one short idea per sticky note.
2. Idea Generation - Students respond to the four prompts:
    - What is your dream vacation destination?
    - If you had one bitcoin, what would you do?
    - What could turn a good day into a bad one?
    - Who is your hero?
3. Circulate and ensure responses are concise and readable.
4. Distribute the three-column handout. Students place their four sticky notes into Beginning, Middle, or End. They may place more than one note in a column.

5. Draft Story - Students construct a short story using all four sticky notes. Requirements:
    - Clear beginning
    - A change or problem in the middle
    - An ending that shows what happened
6. Students form groups of 3–4. Each student tells their story. Listeners give one short comment about clarity or interest.
7. Each student removes one sticky note and passes it to the left. Students reorganize their timeline to include the new idea. Students revise and retell their story.

Optional: Repeat swap once more if time allows.

Checkpoint

1. What changed in your story?
2. Was it easier or harder to include the new idea?

Production

Option 1Students write a final draft (older learners) or draw a three-panel version with captions (younger learners).

Option 2Collect sticky notes and randomly build a class story arc on the board. In small groups, students create a 2-minute role play using all board elements. After performances, briefly identify the turning point and ending in each.

FOLLOW-UP

2 minutes

1. Review and consolidate learning.
2. Ask:
    - Why does order matter in a story?
    - What makes a middle strong?
    - What changed from your first version to your final version?

CLOSE

- Collect sticky notes and handouts.
- Clear board and materials.
- Return markers.
- Reset classroom layout.
- Store or submit student work appropriately.

NOTES

Classroom management

- Distribute sticky notes only at the moment of writing.
- Set visible time limits for each phase.
- Keep group sizes small to ensure participation.

Extensions & Sponge Activities

- Require dialogue in the final draft.
- Add a secret object that must appear in the middle.
- Introduce a rule that the hero cannot appear in the beginning.

Differentiation

- Younger students: Allow drawing before writing, Provide sentence starters.
- ELL/Accessibility: Provide vocabulary support, Allow oral storytelling before written production.
- Safety: Keep movement controlled during role play.

Appendices

- **Appendix A - Three-column student handout template**

6.3 It's Mine!

Activity type Simulation • Memory Game • Team Competition

Duration 45 minutes

Grouping Small Teams (3–4 students) + Whole Class moments

Description

In this activity, learners simulate Bitcoin mining using playing cards as hashes and vocabulary words as transactions. Teams flip cards to find matching pairs. When they find two identical cards, they say “It’s mine.” A valid pair gives them access to the mempool, where they collect one Bitcoin vocabulary word. Words are grouped into blocks and only count after confirmation. The number of cards in the mining pool represents difficulty and is adjusted between rounds. Final scores are calculated by counting the total number of letters in confirmed words stored in each team’s wallet.

Learning Outcomes

By the end of this activity, learners will:

- Explain that mining requires repeated attempts
- Describe how difficulty can increase or decrease

- Understand that transactions are grouped into blocks
- Recognize that confirmation comes before ownership
- Connect effort to reward

Materials

- 1–2 decks of playing cards
- 40–60 Bitcoin vocabulary word cards (*See Appendix A*)
- 1 envelope per team labeled WALLET
- 1 large envelope labeled BLOCK
- Signs labeled MEMPOOL and MINING POOL
- **Nice-to-have:** Stickers or stamps for confirmation, Whiteboard for recording scores

Equipment

- Timer
- Table or floor space for laying out cards
- **Nice-to-have**: Bell or signal sound for block completion

PROCEDURE

OPENING

1. Review key terms briefly: hash, mine, block, wallet, difficulty.
2. Keep explanations short and concrete.
3. Explain that today they must “prove work” before earning words.

PRE-ACTIVITY

5 minutes

1. Lay out 20 playing cards face down in a grid. Explain: Two identical cards = valid hash.
2. When you find a match, say clearly: “It’s mine.”
3. Demonstrate flipping two cards and replacing them if they do not match.

Objective In Student’s Words

- “I find a matching pair and say ‘It’s mine’ to earn a word for my wallet.”

ACTIVITY

30 minutes

Setup

1. Divide students into teams of 3–4.

2. Give each team one WALLET envelope.
3. Place all word cards face down in the MEMPOOL area.
4. Place the BLOCK envelope at the front.

Carry Out

1. Use 20 cards in the mining pool.
2. Teams take turns flipping two cards.
3. If they match, they say “It’s mine.”
4. They collect one word from the mempool.
5. They must read it correctly to keep it temporarily.
6. After 6–8 minutes, stop the round.
7. If too many words were mined, increase difficulty by adding more cards.If too few were mined, decrease difficulty by removing cards.
8. State simply:“When mining is too fast, we make it harder.When mining is too slow, we make it easier.”

Carry Out

1. At the end of each round, collect all earned words into the BLOCK envelope.
2. Announce “Block complete.”
3. Return confirmed words to each team’s wallet.
4. Only confirmed words count.

FOLLOW-UP

4 minutes

- What happened when we added more cards?
- Was it harder to find pairs?
- Did you earn fewer words?
- Did longer words give higher scores?
- Did everyone follow the same rules?

CLOSE

- Collect and count all playing cards and return them to full decks.
- Gather all word cards, empty wallets and the block envelope, and restack neatly.
- Clear the mining area, erase the board, and check for loose cards under tables and chairs.

NOTES

Classroom management

- Only one team flips at a time
- No touching cards out of turn

- Words must be read aloud clearly
- Cards must be returned neatly if unmatched

Extensions & Sponge Activities

- Introduce a halving event: after a set time, each valid hash earns only every second word
- Require red card pairs only for valid hashes
- Add a bonus if a team mines three pairs in one round

Differentiation

- Younger students: Using the word cards directly in the mining pool may make things easier.
- ELL/Accessibility: Add picture support to vocabulary cards. Allow peer reading assistance.
- Safety: Clear space around the mining pool. No running

Appendices

- **Appendix A - 40–60 Bitcoin vocabulary word cards**


My
First
Bitcoin